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# Leitmotif: Understanding Musical Context by Reading a Song as a Graph and as Words

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## Abstract

Musical context lives both in how a song is organised over time and in how people describe it. We present *Leitmotif*, a system that turns each audio clip into a graph of short segments, where edges join neighbouring moments and moments that sound alike, and combines a graph neural network (GNN) over that structure with BERT over captions and tags. Following a four-stage roadmap, we (1) fine-tune BERT for 50-tag multi-label prediction on MusicCaps captions, (2) classify FMA-small genres with GraphSAGE and GAT encoders against a CNN on log-mel spectrograms, (3) fuse the two modalities with bidirectional cross-attention and ablate the fusion strategy, and (4) train a contrastive dual encoder for text-to-music retrieval and zero-shot tagging. We show that the caption-to-tag task leaks labels: 65.6% of positive tag occurrences are spelled out in the caption, and hiding those words lowers BERT's test macro-F1 from 0.659 to 0.458. On FMA-small the best graph model reaches 0.400 macro-F1 against 0.537 for a CNN on spectrograms, and letting a small CNN read each segment's spectrogram patch lifts the graph model to 0.485. Fusing the audio graph with masked captions does not beat text alone (0.448 against 0.468), but it raises average precision for the tags masking hides, such as female vocals (+0.16); and the dual encoder retrieves the matching clip in its top 10 for 8.7% of test captions, 22 times chance. Code, graphs and results are available at <https://github.com/404mahdi/leitmotif>.

## 1 Introduction

Music carries context on several levels at once: genre and era, mood, instrumentation, production quality and form. Some of this context is local, such as the timbre of a distorted guitar, but much of it is relational. A chorus that returns, a riff that loops and a bridge that breaks the pattern are relations between moments of a song rather than properties of any single moment. Convolutional models over spectrograms capture local patterns well [Choi et al., 2016], yet they see such structure only indirectly. Natural-language descriptions summarise context explicitly, but they do not say *where* in the audio it happens.

Leitmotif combines both views. Every clip becomes a graph whose nodes are overlapping one-second segments and whose edges link segments that are adjacent in time or acoustically similar, so repetition turns into explicit connectivity, much like the self-similarity view of music structure [Foote, 1999]. A GNN [Hamilton et al., 2017, Veličković et al., 2018] encodes this graph, BERT [Devlin et al., 2019] encodes captions, and attention [Vaswani et al., 2017] lets each modality query the other. The project follows the four tasks of the course brief: tags from text with a BERT multi-label classifier (Section 7.1); genre from structure with GraphSAGE and GAT encoders on segment and chord graphs, against a CNN on log-mel spectrograms (Section 7.2); fusion of the two modalities for 50 context tags, ablating BERT-only, GNN-only, early concatenation and cross-attention

(Section 7.3); and a contrastive dual encoder for text-to-music retrieval and zero-shot tagging (Section 7.4).

Our contributions are a reproducible pipeline that downloads the data, builds segment and chord graphs and releases sample graphs; an analysis of label leakage in the MusicCaps caption-to-tag task with a masked-caption protocol that removes the most direct leak; a comparison of graph encoders against a CNN under FMA’s artist-filtered split, including a hybrid whose nodes are learned from spectrogram patches; a bidirectional cross-attention fusion whose attention exposes which moments of a clip a caption refers to; and a contrastive text-music model with zero-shot tagging and an automatic relevance study of its results.

## 2 Related work

**Music tagging and structure.** CNNs on mel-spectrograms are the standard supervised approach to music auto-tagging [Choi et al., 2016]. Self-similarity matrices, which compare every moment of a recording with every other, have long been used to reveal repetition and form [Foote, 1999]; our segment graph is a sparsified, learnable version of that idea.

**Graph neural networks.** GraphSAGE aggregates neighbour features with a learned function of the node’s own state [Hamilton et al., 2017], and graph attention networks weight neighbours by learned attention [Veličković et al., 2018]; we use the GATv2 variant, whose attention is a strictly more expressive function of both endpoints [Brody et al., 2022], implemented with PyTorch Geometric [Fey and Lenssen, 2019].

**Language and audio-language models.** BERT provides contextual token representations that transfer well to classification [Devlin et al., 2019]; we use the Hugging Face implementation [Wolf et al., 2020]. Contrastive learning with the InfoNCE objective [van den Oord et al., 2018] aligns paired modalities in a shared space, as in CLIP for images [Radford et al., 2021] and CLAP for audio [Elizalde et al., 2023]. MusicCaps was released alongside MusicLM as an evaluation set of expert music captions [Agostinelli et al., 2023], and FMA is an openly licensed dataset for music analysis with artist-filtered splits [Defferrard et al., 2017].

## 3 Data

**MusicCaps.** MusicCaps contains 5,521 ten-second clips from AudioSet, each with a free-text caption and an aspect list written by a musician. Only YouTube identifiers are distributed, so we downloaded the audio with `yt-dlp`. Cutting the clip by stream copy snaps to container chunk boundaries and returned audio up to 10 s away from the labelled window, so we re-encode at the cut points; the resulting clips match an independent ffmpeg decode with correlation 0.998. Of the 5,521 clips, 4,910 (88.9%) could be retrieved; the others are private, removed or repeatedly failed to download.

MusicCaps has no official split. We use the 2,858 clips flagged as AudioSet evaluation clips as the test set and send 10% of the remaining 2,663 clips to validation using a stable hash of the video id, giving 2,406 / 257 / 2,858 training, validation and test captions. Text-only experiments (Section 7.1) use every caption; experiments that need audio use the clips we could retrieve in each split.

**Tag vocabulary.** The aspect lists contain 13,219 distinct lower-cased phrases, many of them rewordings of each other (*male voice*, *male vocal*, *male singer*). A hand-written synonym table merges such phrases into canonical tags; a phrase may feed several tags (*groovy bass line* counts as *bass* and *groovy*), and unlisted phrases are kept as they are. The 50 most frequent canonical tags on the training split form the label set (Appendix A). They cover genre, mood, instrumentation, voice, tempo and recording quality; the rarest occurs in 101 clips, 95.2% of clips carry at least one, and a clip has 3.89 labels on average.

**Label leakage in captions.** Captions and aspect lists were written by the same annotator at the same time, so a text model can often read a label straight off the caption. Among the 50 most frequent raw aspects, 65.6% of positive clip-aspect pairs appear verbatim in the caption. We therefore also

Table 1: Data used in each stage. MusicCaps clip counts for stages 3 and 4 exclude videos that could not be retrieved.

| Stage             | Input                       | Labels                | Split (train / val / test) | Clips |
|-------------------|-----------------------------|-----------------------|----------------------------|-------|
| 1: tags from text | MusicCaps captions          | 50 tags (multi-label) | 2,406 / 257 / 2,858        | 5,521 |
| 2: genre          | FMA-small audio graphs      | 8 genres (single)     | 6,394 / 800 / 800          | 7,994 |
| 3: fusion         | MusicCaps graphs + captions | 50 tags (multi-label) | 2,121 / 234 / 2,555        | 4,910 |
| 4: retrieval      | MusicCaps graphs + captions | caption pairs         | 2,121 / 234 / 2,555        | 4,910 |

evaluate a *masked* setting in which every phrase that maps to a vocabulary tag is replaced by [MASK], hiding 10.4% of caption words on average: “*The low quality recording features ... mellow piano melody and soft female vocal singing*” becomes “*The [MASK] features ... [MASK] [MASK] melody and [MASK] [MASK] singing*”. Paraphrases such as *poor audio-quality* survive masking, so the masked setting reduces rather than eliminates leakage.

**FMA-small and datasets not used.** FMA-small [Defferrard et al., 2017] contains 8,000 thirty-second MP3 excerpts from eight balanced genres (Electronic, Experimental, Folk, Hip-Hop, Instrumental, International, Pop, Rock; 1,000 tracks each) with an official 6,400 / 800 / 800 split that is filtered by artist; we verified that no artist appears in more than one split. Six training files that are truncated in the FMA release could not be decoded, which leaves 7,994 tracks (6,394 / 800 / 800). We did not use GTZAN, whose repetitions, mislabellings and distortions are well documented [Sturm, 2013] and which has no artist-filtered split. DEAM provides valence and arousal targets, but its songs do not coincide with the clips we have captions for, so the emotion terms of the multi-task loss in the brief are omitted.

## 4 From audio to graphs

**Features.** Audio is resampled to 22,050 Hz mono. With a 2,048-sample window and a 512-sample hop (43.1 frames per second), we compute a 128-bin log-mel spectrogram  $M$ , standardised over the whole clip; 12-bin constant-Q chroma  $C$ ; and 20 MFCCs  $F$  [McFee et al., 2015]. The brief suggests 5–10 s segments, but for 10 s and 30 s clips these give graphs of one to six nodes, too few for message passing. We instead use overlapping 1 s windows with a 0.5 s hop, which gives 18 segments per MusicCaps clip and 57 per FMA track. For a segment  $s_i$  covering frames  $[a_i, b_i]$ , the node feature vector is

$$x_i = [\overline{M}_i; \overline{C}_i; \overline{F}_i; \sigma(F)_i] \in \mathbb{R}^{180}, \quad (1)$$

where bars denote means and  $\sigma$  the standard deviation over the segment’s frames. Node features are standardised with statistics of the training graphs.

**Segment graph.** Temporal edges connect  $s_i$  and  $s_{i+1}$  in both directions. For similarity edges we centre the segment chroma vectors on the clip mean,  $\hat{c}_i = (c_i - \bar{c}) / \|c_i - \bar{c}\|$ , so that similarity reflects what distinguishes two moments rather than what the whole clip shares, and compute  $S_{ij} = \hat{c}_i^\top \hat{c}_j$ . Each segment is linked to at most  $k = 8$  of its most similar segments with  $S_{ij} > \tau = 0.8$ , excluding itself and the overlapping neighbours  $|i - j| \leq 1$ ; the edge set is then symmetrised:

$$\mathcal{E} = \{(i, i \pm 1)\} \cup \{(i, j), (j, i) : j \in \text{top-}k_i(S_i), |i - j| > 1, S_{ij} > \tau\}. \quad (2)$$

Each edge carries  $e_{ij} = [\mathbf{1}_{\text{temporal}}, \mathbf{1}_{\text{similar}}, S_{ij}] \in \mathbb{R}^3$ . On the clips we inspected,  $\tau = 0.8$  keeps roughly the top 5% of centred chroma similarities. Figure 1 shows a segment graph as an arc diagram.

**Chord graph.** For the chord-transition graph, segments follow detected beats. Each beat’s mean chroma is matched against unit-norm templates  $t_c$  of the 24 major and minor triads, and labelled  $\ell_i = \arg \max_c \langle c_i / \|c_i\|, t_c \rangle$  when that score is at least 0.6 and  $N$  (no chord) otherwise; a flat chroma vector scores 0.5 against every triad, so the threshold keeps noise and drums unlabelled. Nodes are the distinct chords in the clip, directed edges join consecutive different chords, and the edge weight counts how often that transition occurs. A node’s features are its chord identity (25-way one-hot), the share of the clip it occupies, and the mean of the 180 segment features over the beats where it sounds (206 dimensions in total).

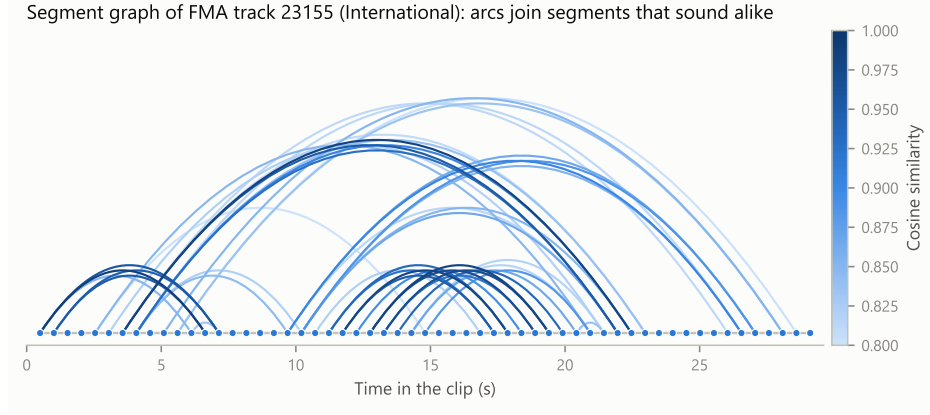


Figure 1: A segment graph drawn as an arc diagram. Dots are one-second segments along the time axis, the grey line is the chain of temporal edges, and each arc is a similarity edge coloured by centred chroma cosine similarity. Long arcs join moments far apart in time that sound alike, which is how repetition appears to the GNN.

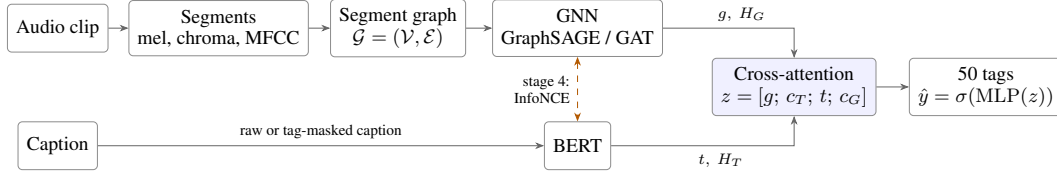


Figure 2: Leitmotif. Audio becomes a segment graph encoded by a GNN, and captions are encoded by BERT. Stage 3 fuses the two with bidirectional cross-attention to predict tags; stage 4 instead aligns the graph embedding  $g$  and the caption vector  $t$  in a shared space trained with InfoNCE.

## 5 Models

**Stage 1: BERT tagger.** A caption  $X$  is tokenised to at most 128 WordPiece tokens and encoded by `bert-base-uncased`. The final [CLS] vector  $t$  feeds one logistic output per tag,  $\hat{y}_k = \sigma(w_k^\top t + b_k)$ , trained with the mean binary cross-entropy over the  $K = 50$  tags,

$$\mathcal{L}_{\text{BERT}} = -\frac{1}{K} \sum_{k=1}^K [y_k \log \hat{y}_k + (1 - y_k) \log(1 - \hat{y}_k)]. \quad (3)$$

**Stage 2: graph encoder.** Node features are first projected to  $d_G = 256$  dimensions,  $h_i^{(0)} = W_0 x_i$ , followed by three message-passing layers, each with a residual connection and layer normalisation. The GraphSAGE layer uses mean aggregation,

$$h_i^{(l+1)} = \text{LN}\left(h_i^{(l)} + \text{Drop}\left(\text{ReLU}\left(W_1^{(l)} h_i^{(l)} + W_2^{(l)} \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} h_j^{(l)}\right)\right)\right), \quad (4)$$

where  $W_1 h_i + W_2 \bar{h}_{\mathcal{N}(i)}$  equals  $W [h_i; \bar{h}_{\mathcal{N}(i)}]$  from the brief with  $W = [W_1 \ W_2]$ . The GAT variant replaces the aggregation with four-head GATv2 attention that also sees the edge attributes,  $\alpha_{ij} = \text{softmax}_j(a^\top \text{LeakyReLU}(\Theta_s h_i + \Theta_t h_j + \Theta_e e_{ij}))$  [Brody et al., 2022]. The graph embedding is the mean of the final node states,  $g = \frac{1}{|V|} \sum_i h_i^{(L)}$ . FMA genres are single-label, so the genre head is a softmax trained with cross-entropy rather than the per-class sigmoid used for tags. We also test two variations. *Crops*: each epoch, every training graph is replaced by a random run of consecutive segments covering half the clip, the graph counterpart of cropping a spectrogram in time, while evaluation uses whole clips. *CNN node features*: a small CNN (three blocks of  $3 \times 3$  convolution with 16, 32 and 64 channels, batch normalisation, ReLU and  $2 \times 2$  max pooling, followed by global mean and max pooling and a linear layer) embeds each segment’s  $128 \times 43$  log-mel patch into 128 dimensions; the embedding is concatenated with  $x_i$  before the input projection and trained end to end with the GNN.

**CNN baseline.** The spectrogram baseline is a VGG-style CNN with four blocks of two  $3 \times 3$  convolutions (32, 64, 128 and 256 channels, batch normalisation, ReLU) and  $2 \times 2$  max pooling, followed by global mean and max pooling, dropout and a linear layer. It receives the full 128-band log-mel spectrogram and neither the graph nor any text.

**Stage 3: fusion.** Let  $H_T \in \mathbb{R}^{L \times d}$  be BERT’s token states projected to  $d = 256$ , with  $t = H_T[\text{CLS}]$ , and let  $H_G \in \mathbb{R}^{N \times d}$  be the projected node states of a clip, with projected graph embedding  $g$ . The *cross-attention* model builds

$$c_T = \text{MHA}(Q = g, K = V = H_T), \quad A = \text{softmax}\left(\frac{(gW_Q)(H_TW_K)^\top}{\sqrt{d_h}}\right), \quad (5)$$

$$c_G = \frac{1}{L} \sum_{l=1}^L \text{MHA}(Q = H_T[l], K = V = H_G), \quad z = [g; c_T; t; c_G] \in \mathbb{R}^{4d}, \quad (6)$$

with four heads of size  $d_h = 64$  and padding positions masked. The first term is the brief’s graph-to-text attention. The second runs from every caption token to the clip’s segments; its attention matrix  $A^{T \rightarrow G} \in \mathbb{R}^{L \times N}$  shows which moments of the audio each word lines up with, which we use for the case studies. A two-layer MLP maps  $z$  to 50 tag logits trained with binary cross-entropy. The ablation replaces  $z$  by  $t$  (BERT only),  $g$  (GNN only) or  $[g; t]$  (early concatenation). In every variant the top four BERT layers are fine-tuned and the rest stay frozen, and the graph encoder is the stage-2 GAT with CNN node features, initialised from its FMA-small weights; stage 4 uses the same initialisation.

**Stage 4: contrastive dual encoder.** Two-layer projection heads map  $g_i$  and  $t_i$  to unit vectors  $\hat{g}_i, \hat{t}_i \in \mathbb{R}^{256}$ . With in-batch negatives and a learnable temperature  $\tau$  (initialised to 0.07,  $1/\tau \leq 100$ ), the symmetric InfoNCE loss [van den Oord et al., 2018, Radford et al., 2021] over a batch of  $N$  pairs is

$$\mathcal{L}_{\text{NCE}} = -\frac{1}{2N} \sum_{i=1}^N \left[ \log \frac{\exp(\hat{g}_i^\top \hat{t}_i / \tau)}{\sum_j \exp(\hat{g}_i^\top \hat{t}_j / \tau)} + \log \frac{\exp(\hat{g}_i^\top \hat{t}_i / \tau)}{\sum_j \exp(\hat{g}_j^\top \hat{t}_i / \tau)} \right]. \quad (7)$$

For zero-shot tagging, each tag  $k$  is embedded as the normalised mean of the text embeddings of three prompts (“{tag}”, “this music is {tag}”, “a recording featuring {tag}”), and clip  $i$  receives the score  $\hat{g}_i^\top u_k$ .

## 6 Experimental setup

All models are trained with AdamW [Loshchilov and Hutter, 2019], a linear warm-up over the first 10% of steps (5% for models without BERT) followed by linear decay, bfloat16 mixed precision and gradient clipping at 1.0. Pretrained BERT layers use a small learning rate and newly initialised layers a larger one; Appendix C lists every hyperparameter. We keep the checkpoint with the best validation macro-F1 (mean recall for stage 4) and report test results for that checkpoint only. For tag prediction, a single decision threshold is chosen on validation from  $\{0.05, 0.10, \dots, 0.95\}$  to maximise macro-F1 and then applied to the test set; per-epoch curves use 0.5. Training ran on a single NVIDIA RTX 5060 Laptop GPU with 8 GB of memory: a stage-1 epoch takes about 15 s, a GNN epoch on FMA-small 2–4 s, a CNN epoch about 55 s and a stage-3 epoch 10–15 s. The random seed is fixed at 42.

**Hyperparameter rounds.** Stages 3 and 4 were each run with two settings, and the choice between them used validation data only. Stage 3 first ran for 15 epochs with a single learning rate for the pretrained graph encoder and for every newly initialised layer; all variants were still improving at the final epoch, so a second round used 25 epochs and a five times smaller learning rate ( $2 \times 10^{-4}$ ) for the graph encoder. Validation macro-F1 improved for every variant that uses text (BERT only 0.518 to 0.548, early concatenation 0.510 to 0.534, cross-attention 0.493 to 0.495) and the ordering did not change, so the second round is the one reported. Stage 4 was run with four and with two trainable BERT layers; two layers was worse on validation (mean recall 0.337 against 0.350), so the four-layer setting is reported. Test numbers were computed once, for the selected setting of each stage.

Table 2: Stage 1, MusicCaps test split (2,858 captions, 50 tags). The threshold is chosen on validation.

| Model                     | Macro-F1     | Micro-F1     | AUC-PR       | Threshold |
|---------------------------|--------------|--------------|--------------|-----------|
| B1: random, tag frequency | 0.078        | 0.125        | 0.078        | —         |
| BERT, captions as written | <b>0.659</b> | <b>0.720</b> | <b>0.680</b> | 0.40      |
| BERT, tag words masked    | 0.458        | 0.569        | 0.471        | 0.30      |



Figure 3: Stage 1 macro-F1 and micro-F1 by epoch at threshold 0.5, on raw (left) and masked (right) captions.

**Metrics and baselines.** For tags we report macro-F1, micro-F1 and AUC-PR (the mean over tags of average precision, skipping tags without positives); for genre, accuracy, macro-F1 and one-vs-rest AUC-PR; for retrieval, recall at  $K \in \{1, 5, 10\}$  and the median rank of the true match in both directions over the whole test split. Baseline B1 guesses each tag at random with its training-set frequency, or always predicts the most common training genre; B2 is the CNN on log-mel spectrograms; B3 is the BERT-only model of stage 1 and its audio-subset counterpart in the stage-3 ablation.

## 7 Results

### 7.1 Stage 1: tags from text

Table 2 shows that BERT is far above random guessing on both versions of the captions. Masking the tag words costs 0.20 macro-F1 and 0.21 AUC-PR, which confirms that a large part of the raw-caption score comes from reading labels directly off the text. The masked model still reaches 0.458 macro-F1, so the surrounding words (“it sounds like something you would hear at Sunday services”) carry real context. Figure 3 shows that both runs converge within ten epochs and that the gap between validation and test is larger in the masked setting.

### 7.2 Stage 2: genre from structure

All graph models are about three times as accurate as the majority baseline, and segment graphs beat chord graphs for both encoders: similarity of timbre and harmony over one-second windows says more about genre than a sequence of template-matched triads, which are unreliable for genres such as Experimental or Hip-Hop that are not built on chord progressions. GAT, which also sees the edge attributes, is slightly better than GraphSAGE on both graph types. The CNN on full log-mel spectrograms is nevertheless clearly stronger (0.537 against 0.400 macro-F1). We attribute most of the gap to the node features, which summarise a second of audio by its mean spectrum, mean chroma

Table 3: Stage 2, FMA-small test split (800 tracks, 8 balanced genres). “+ crops” trains on random half-length crops and evaluates on whole clips; “+ CNN node features” adds a learned embedding of each segment’s log-mel patch to its node features.

| Model                                  | Accuracy     | Macro-F1     | AUC-PR       |
|----------------------------------------|--------------|--------------|--------------|
| B1: majority genre                     | 0.125        | 0.028        | 0.125        |
| B2: CNN on log-mel                     | <b>0.541</b> | <b>0.537</b> | <b>0.572</b> |
| GraphSAGE, chord graph                 | 0.364        | 0.356        | 0.393        |
| GAT, chord graph                       | 0.369        | 0.372        | 0.396        |
| GraphSAGE, segment graph               | 0.384        | 0.379        | 0.403        |
| GAT, segment graph                     | 0.396        | 0.400        | 0.411        |
| GraphSAGE, segment graph + crops       | 0.374        | 0.369        | 0.375        |
| GAT, segment graph + crops             | 0.378        | 0.378        | 0.388        |
| GAT, segment graph + CNN node features | 0.484        | 0.485        | 0.514        |

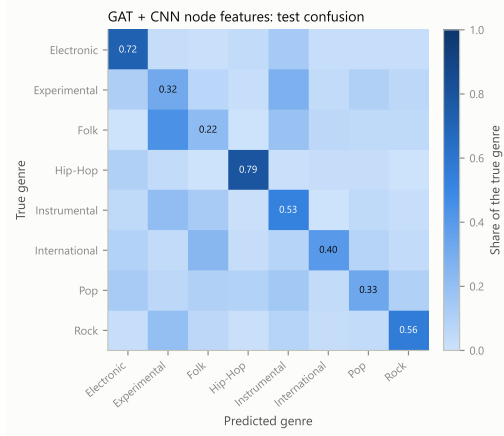


Figure 4: Test confusion matrix of the best stage-2 graph model, normalised by true genre. Diagonal values are per-genre recall.

and a few MFCC statistics and so discard the fine time–frequency texture the CNN learns from. The graph models also overfit, reaching near-zero training loss within 60 epochs while validation macro-F1 plateaus around 0.45, and random half-length crops did not help (0.369 and 0.378).

Giving the graph model that texture back closes most of the gap: with a small CNN embedding each segment’s spectrogram patch, the same GAT rises from 0.400 to 0.485. The hybrid still trails the CNN overall, but its errors differ (Figure 4): it recognises Electronic (recall 0.72 against 0.58) and Rock (0.56 against 0.50) better, while the CNN leads on International (0.68 against 0.40) and Pop (0.52 against 0.33). Loop-based electronic music and riff-driven rock plausibly expose the repetition the graph encodes, though we did not test that directly. As the strongest graph encoder, the hybrid initialises the graph side of stages 3 and 4.

### 7.3 Stage 3: fusing structure and text

The audio graph on its own reaches 0.227 macro-F1, about three times the random baseline of 0.078, so a clip’s structure and texture do carry usable context. Text is much stronger, and neither fusion improves on it: early concatenation gives up 0.02 macro-F1 against BERT alone and cross-attention 0.03. The ordering held under both hyperparameter settings we tried (Section 6), so it is not an artefact of one schedule.

The aggregate hides what fusion actually does. Figure 7 in Appendix D compares per-tag average precision of the strongest fusion variant with BERT alone. Audio helps most where a masked caption says least about the sound: *female vocals* (+0.16 AP), *exciting* (+0.10), *male vocals* (+0.08) and *energetic* (+0.07). It costs accuracy on tags that the surrounding words still imply, such as *mellow*

Table 4: Stage 3 ablation on the 2,555 MusicCaps test clips with audio, captions with tag words masked, 50 tags. Every variant that uses audio starts from the stage-2 GAT with CNN node features.

| Fusion                            | Macro-F1     | Micro-F1     | AUC-PR       |
|-----------------------------------|--------------|--------------|--------------|
| GNN only (audio graph)            | 0.227        | 0.304        | 0.170        |
| BERT only (masked caption)        | <b>0.468</b> | <b>0.556</b> | <b>0.459</b> |
| Early concatenation [ $g$ ; $t$ ] | 0.448        | 0.535        | 0.438        |
| Cross-attention                   | 0.440        | 0.528        | 0.428        |

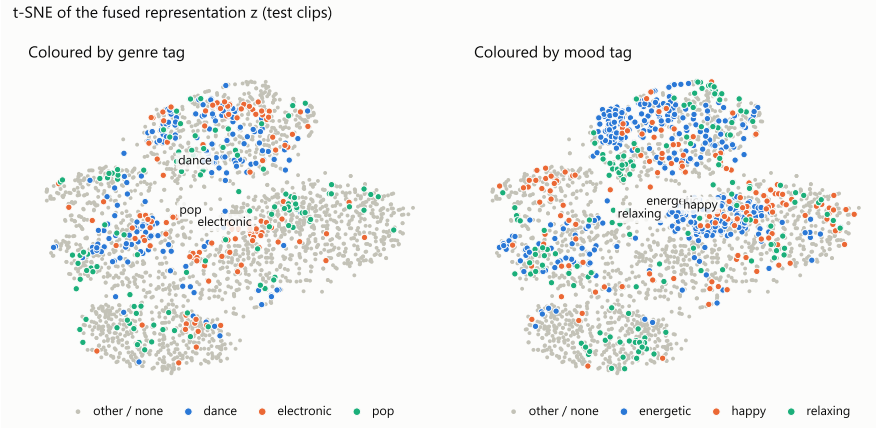


Figure 5: t-SNE of the fused representation  $z$  for test clips, coloured by the most frequent genre tags (left) and mood tags (right). Clips with none of the three tags are grey.

( $-0.12$ ), *live performance* ( $-0.12$ ) and *eerie* ( $-0.12$ ). Only 16 of the 50 tags improve with early concatenation and 11 with cross-attention, which is why the macro average falls. Both fusions mix the two modalities the same way for every tag; a model that could route each tag to whichever modality predicts it better would not have to make that trade.

Figure 5 projects the fused representation  $z$  of the test clips with t-SNE. Mood separates more clearly than genre: *energetic* and *relaxing* clips sit in different regions, while *dance*, *electronic* and *pop* overlap, which matches the confusion between those genres in stage 2.

The case studies in Figure 6 show what the cross-attention actually learns. Its text-to-segment attention is not word-specific: within a clip, nearly every caption word attends to the same one or two segments, which appear as vertical stripes in the attention maps. The model therefore learns *where* the informative audio is, for example the last second of a harp recording or the chorus entry of a live performance, rather than aligning individual words with moments. That is a useful signal for locating the part of a clip a description refers to, but it is weaker than the word-level alignment the design allows for.

#### 7.4 Stage 4: text-to-music retrieval

Retrieval is far above chance and still weak in absolute terms.  $R@10$  of 0.087 is 22 times the 0.004 a random ranking would give, and the matching clip has a median rank of 151 out of 2,555, the top 6%. Both directions behave the same way, as the symmetric loss intends. The model memorises its training pairs: the training loss falls from 4.11 to 0.11 while validation mean recall peaks at epoch 10 of 30, which is what 2,121 caption-clip pairs allow. Freezing more of BERT and slowing the graph encoder, the settings that helped stage 3, made validation mean recall slightly worse (0.337 against 0.350), so the first setting was kept.

Zero-shot tagging through the shared space is close to useless: scoring each test clip against prompt embeddings of the 50 tags gives 0.089 mean average precision, barely above the 0.078 of random ranking, and far below the supervised models of stage 3 (0.170 for the graph alone, 0.459 for BERT on masked captions). Retrieval trains the space to separate whole captions from

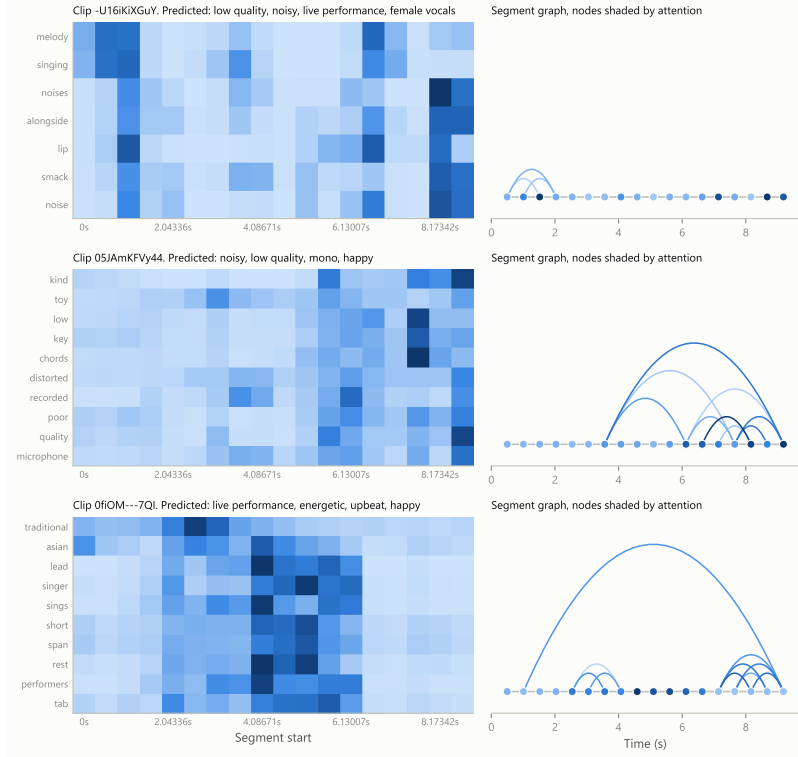


Figure 6: Case studies. Left: caption words (rows) against one-second segments (columns) under the text-to-graph attention  $A^{T \rightarrow G}$  of the cross-attention model. Right: the same clip’s segment graph, with nodes shaded by the attention they receive averaged over the words shown.

Table 5: Stage 4 retrieval over the 2,555 MusicCaps test clips with audio. Chance R@10 is 0.004.

| Direction                   | R@1   | R@5   | R@10  | Median rank |
|-----------------------------|-------|-------|-------|-------------|
| Caption $\rightarrow$ audio | 0.013 | 0.051 | 0.087 | 151         |
| Audio $\rightarrow$ caption | 0.013 | 0.056 | 0.095 | 140         |

each other, not to place short tag phrases where the matching audio lies, and 2,121 pairs are too few to learn that alignment. Ten caption queries with their three closest clips are saved in `results/retrieval_examples/examples.json`; they are plausible without being right, as when a query describing a live hip-hop recording with male vocals, piano, bass and hi-hats returns a clip whose own caption reads “an erotic male voice ... sustained soft piano synth and a drum kit”.

**Relevance of retrieved clips.** The brief suggests rating retrieved clips with at least five human listeners. We did not run a listening study. Instead, each of the 30 retrieved clips (10 caption queries, top 3 each) and three random clips per query are scored on a 1–5 scale by three automatic judges that share no weights with our models: tag agreement (1 + 4  $\times$  the Jaccard overlap of the query clip’s and candidate clip’s tags), caption agreement (cosine similarity of the two captions under the all-MiniLM-L6-v2 sentence encoder) and audio–text agreement (cosine similarity of the query caption and the candidate’s audio under the LAION CLAP model `clap-htsat-unfused`). The last two are expressed as 1 + 4*p*, where *p* is the share of random test caption pairs, or caption–clip pairs, that score lower. Retrieved clips beat random clips on all three judges (Table 6): tag overlap 1.70 against 1.32, caption similarity 3.61 against 3.13, and audio–text agreement 4.49 against 3.45, with the clip the caption was actually written for reaching 4.80. Retrieval therefore returns music that is genuinely related to the query, even though the correct clip is rarely first. These scores are a proxy for, not a substitute for, human judgement; CLAP was pretrained on general audio–text data, so it may share biases with our text encoder.

Table 6: Automatic relevance of retrieved clips on a 1–5 scale: 10 caption queries, their three closest clips, and three random test clips per query as a reference. A score of 3 means “as similar as a typical random pair”.

| Clips                          | Tag overlap | Caption similarity | Audio–text agreement |
|--------------------------------|-------------|--------------------|----------------------|
| Rank 1                         | 1.95        | 3.63               | 4.42                 |
| Rank 2                         | 1.37        | 3.33               | 4.46                 |
| Rank 3                         | 1.78        | 3.87               | 4.59                 |
| Retrieved, all ranks           | 1.70        | 3.61               | 4.49                 |
| Random clips                   | 1.32        | 3.13               | 3.45                 |
| The clip the caption describes | –           | –                  | 4.80                 |

## 8 Discussion and limitations

Three findings stand out, and none of them is the one we set out to show.

**Structure carries real information, but node features decide how much.** A segment graph alone predicts genre three times better than the majority class and 50 context tags three times better than random guessing, so repetition and similarity between moments are genuinely informative. Yet with hand-made node summaries the graph model reaches only 0.400 macro-F1 on FMA genres against 0.537 for a CNN on the same audio. Letting a small CNN read each segment’s spectrogram patch, while message passing still runs over the same graph, recovers most of that gap (0.485). What limited the graph model was not the graph but what each node was allowed to see.

**Fusion helps per tag, not on average.** Neither early concatenation nor cross-attention beats BERT on masked captions (0.448 and 0.440 against 0.468). The per-tag view explains why: audio raises average precision for tags the masked caption no longer states, above all *female vocals* (+0.16), *exciting* (+0.10) and *male vocals* (+0.08), and lowers it for tags the remaining words still imply. Only 16 of 50 tags improve, and a macro average over 50 tags turns that trade into a loss. Both fusions combine the modalities identically for every tag; a gate that learned, per tag, how much to trust each side is the obvious next step and is what these numbers argue for.

**Retrieval works, weakly.** The contrastive space puts the matching clip in the top 10 for 8.7% of captions, 22 times chance, with a median rank of 151 out of 2,555, and three independent judges score retrieved clips well above random clips. It also memorises its 2,121 training pairs within ten epochs, and its zero-shot tagging is barely above random, so the space aligns whole captions rather than concepts. More paired audio and text, or an audio encoder pretrained at scale, would matter more here than any change of architecture.

**Limitations.** Captions and tags were written together, and masking removes only exact phrases. The labelled data is small: stages 3 and 4 train on 2,121 clips, and 611 MusicCaps videos are no longer available. Template-matched chords are a coarse estimate of harmony. FMA genres are single-label, and FMA and MusicCaps differ in recording conditions, which limits transfer of the pretrained graph encoder. The cross-attention maps are not word-specific, so the alignment they give is between a caption and a moment, not between a word and a moment. Finally, retrieval quality is judged automatically rather than by listeners.

## 9 Conclusion

We built a pipeline that reads a song as a graph of its own repetitions and as the words people use to describe it, and evaluated it across four stages on FMA-small and MusicCaps. Song structure predicts genre and context tags well above chance, and its usefulness grew sharply once each node could see the spectrogram rather than a summary of it, but a plain CNN still classifies genre better, and text still predicts tags better than text plus audio. The value of the audio graph shows up per tag and per moment: it improves exactly the tags a caption stops describing once its tag words are hidden, and its attention marks where in a clip the informative audio sits. For the tasks in this project, the honest conclusion is that graph structure is a useful complement to language rather than

a replacement for stronger audio features, and that the fusion should be allowed to weigh the two modalities per tag.

**Reproducibility.** The repository contains the full pipeline, from download scripts to training, evaluation and figures, with exact dependency versions (`uv.lock`), the configuration used for every run (`config.yaml`), the data splits, 20 sample graphs in `.pt` and `.json` form, and an end-to-end inference notebook.

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## A Tag vocabulary

The 50 canonical tags, from most to least frequent on the training split: low quality, instrumental, bass, male vocals, passionate, groovy, medium tempo, noisy, energetic, emotional, live performance, amateur recording, slow tempo, fast tempo, electric guitar, mono, happy, piano, female vocals, acoustic drums, percussion, relaxing, shimmering cymbals, punchy drums, keyboard, electronic drums, pop, acoustic guitar, classical, no other instruments, electronic, dance, synth, upbeat, exciting, rock, eerie, romantic, strings, melodic, fun, sad, mellow, folk, easygoing, violin, trumpet, soulful, claps, youthful. Examples of merged phrases: *low quality*  $\leftarrow$  poor audio quality, bad audio quality, low quality recording; *bass*  $\leftarrow$  bass guitar, e-bass, groovy bass line; *relaxing*  $\leftarrow$  calming, soothing, chill, meditative. The full table is in `src/tags.py`.

## B Example stage-1 predictions

Five test clips drawn at random, with the tags each stage-1 model ranks highest. Reading the two rows together shows what masking removes. For the country clip, the raw model puts *male vocals* first at 0.98 because the caption opens with “A male singer sings”; once that phrase is masked the same tag falls to 0.24 and the model leans on *emotional* and *romantic* instead. Where the caption describes the sound without naming a tag, as in the first clip, both models agree. Full records, including the masked captions, are in `results/stage1_examples.json`.

Table 7: Top-5 predicted tags with probabilities, for captions as written and with tag words masked.

| Clip        | True tags                                                         | Raw caption                                                                                  | Masked caption                                                                                    |
|-------------|-------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| 3m4QSVVBMss | bass, claps, groovy, low quality, pop, punchy drums               | low quality 0.99, bass 0.94, groovy 0.93, punchy drums 0.92, pop 0.80                        | low quality 0.99, groovy 0.92, bass 0.86, passionate 0.84, shimmering cymbals 0.83                |
| Nqb7nw58q08 | bass, energetic, exciting, groovy, instrumental, live performance | instrumental 0.97, exciting 0.92, groovy 0.86, bass 0.65, passionate 0.55                    | instrumental 0.96, groovy 0.92, passionate 0.74, bass 0.65, exciting 0.55                         |
| O0sDg-yLvIE | emotional, live performance, male vocals, romantic, slow tempo    | male vocals 0.98, emotional 0.93, slow tempo 0.68, romantic 0.67, acoustic guitar 0.62       | emotional 0.88, romantic 0.71, medium tempo 0.33, female vocals 0.33, male vocals 0.24            |
| ad6UhYwTXXQ | amateur recording, female vocals                                  | amateur recording 0.91, female vocals 0.59, low quality 0.17, slow tempo 0.10, relaxing 0.06 | amateur recording 0.44, classical 0.38, instrumental 0.14, live performance 0.13, fast tempo 0.12 |
| iMmYVLSb1IY | medium tempo                                                      | instrumental 0.99, piano 0.35, slow tempo 0.26, no other instruments 0.24, medium tempo 0.23 | instrumental 0.91, slow tempo 0.68, synth 0.41, eerie 0.27, no other instruments 0.27             |

## C Hyperparameters

Table 8: Training settings. Learning rates are given as pretrained BERT layers / newly initialised layers, and for stage 3 as BERT / pretrained graph encoder / new layers; wd is the AdamW weight decay.

| Model                | Epochs | Batch | Learning rate                                   | Other settings                                              |
|----------------------|--------|-------|-------------------------------------------------|-------------------------------------------------------------|
| Stage 1 BERT tagger  | 10     | 32    | $3 \times 10^{-5} / 10^{-3}$                    | all layers tuned, max length 128, wd 0.01                   |
| Stage 2 GNN          | 60     | 64    | $10^{-3}$                                       | 3 layers, width 256, dropout 0.3, wd $10^{-4}$              |
| Stage 2 + CNN nodes  | 20     | 32    | $10^{-3}$                                       | as above; patch CNN with 16/32/64 channels, 128-d embedding |
| B2 CNN               | 30     | 32    | $10^{-3}$                                       | dropout 0.3, wd $10^{-4}$                                   |
| Stage 3 fusion       | 25     | 32    | $2 \times 10^{-5} / 2 \times 10^{-4} / 10^{-3}$ | $d=256$ , 4 heads, top 4 BERT layers, wd 0.01               |
| Stage 4 dual encoder | 30     | 64    | $2 \times 10^{-5} / 5 \times 10^{-4}$           | $\tau_0=0.07$ , top 4 BERT layers, wd 0.01                  |

## D Additional figures

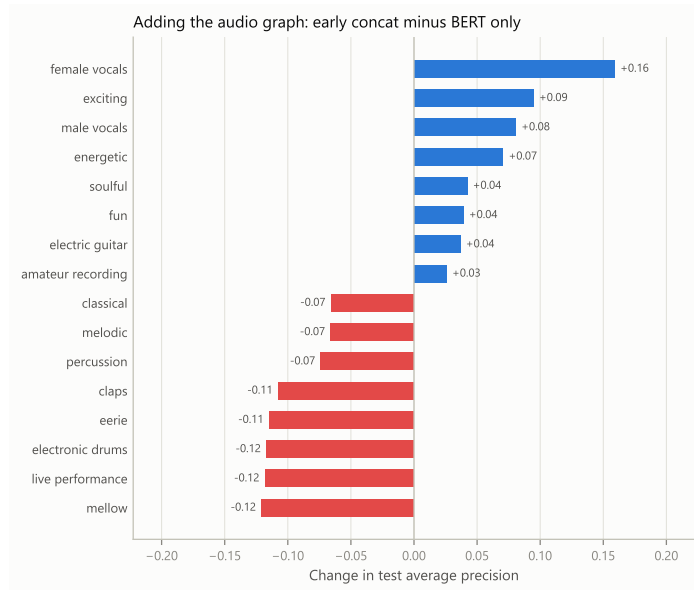


Figure 7: Per-tag change in test average precision when the audio graph is added to the masked caption (early concatenation minus BERT only). Blue tags improve, red tags get worse; the eight largest movements in each direction are shown.

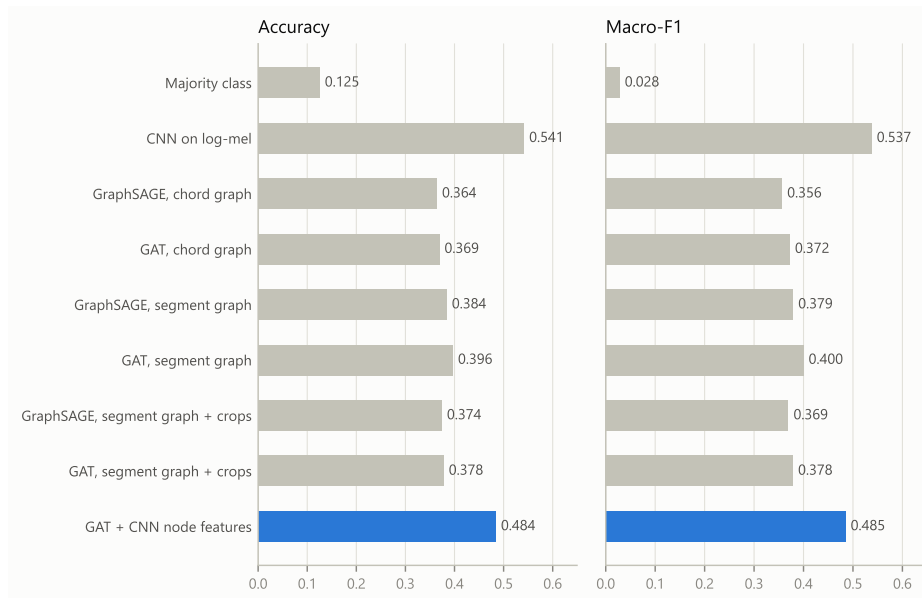


Figure 8: Stage 2 test accuracy and macro-F1 on FMA-small. The best model that is not a baseline is blue; baselines and the other models are grey.

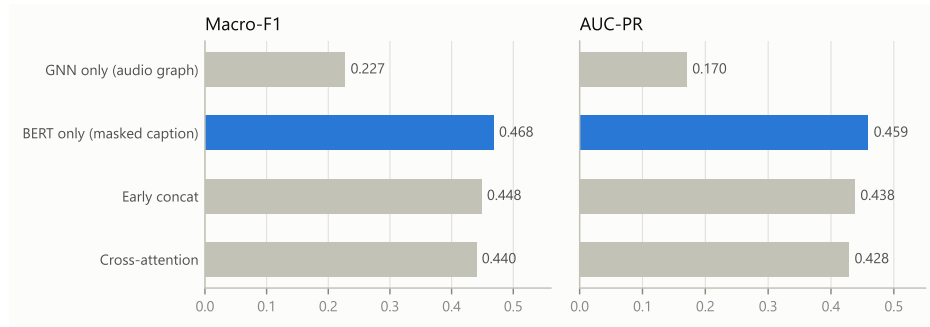


Figure 9: Stage 3 ablation on masked captions: test macro-F1 and AUC-PR. The best fusion variant is blue.

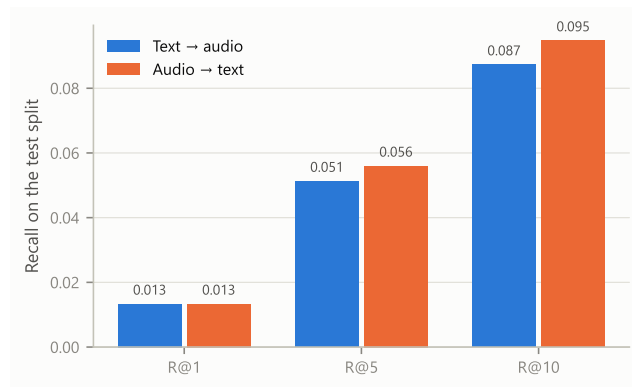


Figure 10: Stage 4 recall at  $K$  over the MusicCaps test clips with audio, for caption-to-audio and audio-to-caption retrieval.